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NPS/002/017 – Technical Specification for 66kV and 132kV Cable Accessories.

1. Purpose

The purpose of this document is to detail the technical requirements of 66kV and 132 kV cable accessories for use on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/017	2.0	Specification for 66 and 132kV Cable accessories including oil transition joints.

2. Scope

This specification details the technical requirements for 66kV and 132 kV cable accessories. This includes XLPE to XLPE, Transition Joints and Terminations for use on the CE Electric UK distribution network.

Service failures have been experienced on terminations, due to longitudinal shrinkage of cable over sheath material and disturbance to the lead sheath, moisture seals and earth connections. All accessories shall, as far as is reasonably practicable, accommodate or prevent movement of cable components occurring as a result of cable sheath retraction.

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3. Technical Requirements

3.1 Conditions of Operation.

Cable accessories purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/913 and IMP/001/914. The following are general conditions of operation:

- Nominal system voltages: 66kV and 132 kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points

Typical cable types and sizes that may require jointing are listed in Appendix 1.

3.2 66kV and 132kV Cable Joints

Cable joints are required to make a connection between cables of various sizes and types. Existing cables typically are:

- Single or three-core paper insulated with copper or aluminium conductors, fluid filled cables, with either lead or aluminium sheaths, and PVC / PE over sheaths.
- Single core triple extruded XLPE insulation, bonded screens, copper conductors, lead sheathed and a PE over sheath to IEC 60840 and BS 7912. XLPE cables may contain water swellable tapes or powders.

3.2.1 Single core joints for XLPE to XLPE

XLPE / XLPE joints will be tested to meet the requirements of IEC 60840 and BS 7912 or equivalent. New installations on XLPE cables at 132kV shall be cable systems as defined in IEC 60840 clause 12, this is also the preferred arrangement for 66kV installations.

3.2.2 Transition Joints (single core XLPE / single core and three core fluid filled cables)

Joints will be tested to meet the requirements of IEC 60840 and BS 7912 or equivalent. Additionally this will include the hydraulic, lightning impulse and dielectric security tests specified in IEC 60141.

3.3 66kV and 132kV Terminations.

Terminations will be made on 66kV or 132kV XLPE cables to IEC 60840 and BS 7912, and tested to meet the requirements of these specifications. New installations on XLPE cables at 132kV shall be cable systems as defined in IEC 60840 clause 12. This is also the preferred arrangement for 66kV installations.

3.3.1 Indoor Terminations

Indoor terminations shall comply with the requirements of IEC 60840 clauses 12 and 14 as applicable, and clause 16 of BS 7912.

3.3.2 Outdoor Terminations

Outdoor terminations shall comply with the requirements for indoor terminations and the specific requirements of clause 16.12 of BS 7912 for outdoor terminations.

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3.3.3 GIS Terminations

GIS terminations shall comply with the requirements for indoor terminations and the specific requirement of clause 16.13 of BS 7912 for GIS terminations and IEC 60859.

3.3.4 Pollution Level and Creepage Requirements

Unless otherwise specified the minimum specific creepage distance (SCD) shall be 25mm/kV as detailed in IEC 60815: Table J.1.

3.4 Earthing - Auxiliary Equipment

New cable systems will normally be “solidly bonded”, with special bonding arrangements applicable to existing installations. Auxiliary equipment required for earthing and earth connections shall comply with the requirements of IEC 60840 and the BS 7912 clause 17.

3.4.1 Bonding leads.

Earth bonding leads shall be appropriate for the required fault level of the cable system to which they are connected, and are normally 120mm² Cu for 66kV and 240mm² Cu for 132kV cable systems respectively. Bonding leads shall comply with the requirements of BS7912 clause 17.1.

3.4.2 Cable Sheath Link boxes.

Cable sheath link boxes shall meet the requirements of NPS/002/026 – Technical Specification for Cable Sheath Link Boxes.

3.4.3 Sheath Voltage Limiters (SVL)

Where the cable earthing and bonding arrangement dictates the use of SVL's, these will be installed in an earthed metal housed cable sheath link box as detailed in section 3.4.2, and can be supplied as single phase or as a three phase unit. Where they are supplied as single phase units they must be a “matched set” with identical characteristics. SVL's shall meet the requirements in BS 7819 clause 17.2.

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4 References

4.1 External Documentation

The joints and terminations described within this specification shall comply with all current versions of the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard.

In respect the following documents are particularly relevant

Reference	Title
Energy Network Association Technical Specification	
ENATS 09-02	Specification for the supply, delivery and installation of power cables with operating voltages in the range 33kV to 400kV and associated auxiliary cables
ENATS 09-04	66kV and 132kV Impregnated paper insulated oil filled and gas pressure type cable systems.
British Standards	
BS7912: 2001	Power cables with XLPE insulation and metallic sheath, and their accessories, for rated voltages from 66 kV (=72.5 kV) to 132kV (=145kV). Requirements and test methods
BS6480: 1998	Specification for impregnated paper-insulated lead or lead alloy sheathed electric cables of rated voltage up to and including 66kV and 132kV
BS7819: 1995	Guidelines for application of analytical test methods for thermal endurance testing of electrical insulating materials
BS7970: 2005	Electric cables. Metallic wire and foil sheath constructions of power cables having XLPE insulation for rated voltages from 66kV (U = 72.5 kV) to 132kV (U = 145 kV)
IEC Technical Specification	
IEC 60840: 2004	Power cables with extruded insulation and their accessories for rated voltages up to 150kV
IEC 60141-1: 1993	Oil filled paper insulated, metal sheathed cables and accessories for alternating voltages up to 400kV.
IEC 60859: 1986	Guide for cable connections for gas-insulated metal-enclosed switchgear for rated voltages of 72.5 kV and above
IEC 60815	Guide for the Selection of Insulators in Respect of Polluted Conditions

4.2 Internal Documentation

Reference	Title
IMP/001/913.	Code of Practice for the Economic Development of the EHV System.
IMP/001/914	Code of Practice for the Economic Development of the 132kV System.

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4.3 Amendments from Previous Version

Clause	Subject	Amendments
2	Scope	Reference made to cable sheath retraction.
3.1 and 4.2	Conditions of Operation & Internal Documents	Reference made to the Code of Practice (CoP) IMP/001/913 and IMP/001/914.
3.2	66kV & 132kV Cable Joints	Redrafted, incorporating section 3.5 (cable joints) of previous version. Amendments made to cable type, joint descriptions and BS / IEC document references. Indicates preferred arrangement for circuits at 66kV as cable systems.
3.2.1	Single core joints for XLPE to XLPE	Redrafted, incorporates parts of section 3.5.1 (XLPE single cores joints) of previous version.
3.2.2	Transition Joints	Redrafted, incorporates parts of section 3.5.1 and 3.5.2 of previous version.
3.3	66kV & 132kV Cable Terminations	Redrafted. Indicates preferred arrangement for circuits at 66kV as cable systems.
3.4	Earthing - Auxiliary Equipment	Redrafted, indicates "normal" requirement for solidly bonded systems.
3.4.2	Cable Sheath Link Boxes	Redrafted (previously section 3.4.3) and refers to NPS/002/026 – Technical Specification for Cable Sheath Link Boxes.
4.1	External Documents	Tables amended and reference to ENATS 09-2 and ENATS 09-3 added.
Appdx 1	Cable Types	Typical cable types amended.

5. Definitions

Term	Definition
NIL	

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6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	19/03/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	29/03/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	22/03/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	14/04/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	31/03/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	31/03/2010

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Appendix 1 – Typical cable types

66kV and 132kV

- 3c 0.25 to 1.0² in (or metric equivalent): Copper or aluminium conductor, fluid filled, screened, lead or aluminium sheath and PVC over sheathed.
- 1c 0.25 to 1.0² in (or metric equivalent): Copper or aluminium conductor, fluid filled, screened, lead or aluminium sheath and PVC over sheathed.
- 1c 185 -1000 mm²: Single Core XLPE: stranded copper conductor, lead sheathed and MDPE over sheathed.

Types of Cable capable of being terminated

The supplier shall provide details in Appendix 4 (self certification conformance declaration) of the cable type that each joint or termination can be used in conjunction with. This shall include the size; build up material types, manufactures reference and place of manufacture.

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Appendix 2 – Addendum to Supplier Requirements

Work Instructions

The supplier shall provide work instructions with each kit.

Kitting

All joints including terminations will be supplied in kit form where all the components required including connectors, lugs and work instructions shall be included within the kit.

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Appendix 3 – Logistical Requirements

Storage

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment, maximum stacking height with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to:

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

Packaging and labelling

The Tenderer shall ensure that each item is suitably packaged and protected to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant, multiple packages of like products. Where product packages tendered are made up of sub packages each sub package shall be marked. As a minimum requirement the following shall be included:

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

Power Cable accessories for rated voltages 66kV to 132kV shall comply with the latest issues of the relevant international and international standards.

IEC 60840: 2004, IEC 60141 – 1:1993 and BS7912: 2001 are intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses of the aforementioned standard relevant to Power Cables Accessories for rated voltages 66kV to 132kV for use on the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

**Details of the cable type(s)
designed to be used with:**

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Appendix 4

Technical Specification for Power Cable Accessories for rated voltages 66kV to 132kV

• IEC 60840: 2004 (Cable Systems)			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
12.1	Range of type approval		
12.3.1	Test voltage values		
12.3.3	Bending test		
12.3.5	Tan δ		
12.3.6	Heating cycle voltage test		
12.3.4	Partial discharge test (ambient and high temp)		
12.3.7	Lightening impulse voltage test followed by power frequency voltage test		
Annex H	Test of outer protection of buried cables		
12.3.8.1	Examination		

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• IEC 60840: 2004 (Accessories)			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
14.1	Range of type approval		
12.3.1	Test voltage values		
12.3.4	Partial discharge test at high temperature		
12.3.6	Heating cycle voltage test		
12.3.4	Partial discharge test (ambient and high temp)		
12.3.7	Lightening impulse voltage test followed by power frequency voltage test		
Annex H	Test of outer protection of buried cables		
12.3.8.1	Examination		

• IEC 60141-1: 1993			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
5.2	Routine pressure test		
7.3	Lightening Impulse Voltage Test		
7.4	Dielectric security test		

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• IEC 60815			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
Table J.1	Specific Creepage Distance (SCD) 25kV/mm		

• BS7912: 2001			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
16.2.1.	Test and requirements – General		
12.5	Bending test		
16.3	Partial discharge test		
16.4	Heating cycle voltage test		
16.3	Partial discharge test		
16.5	Lightening impulse voltage test		
16.6	Power frequency voltage test		
16.7	DC Voltage test		
	Examination		
16.11	Tests on Buried Joints		
16.11.2	Mechanical tests		
16.11.3	Water immersion and heat cycling		
16.11.4	DC voltage withstand		
16.11.5	Impulse withstand		

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• BS7912: 2001			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
16.12	Tests on outdoor terminations		
16.12.1	Salt fog test		
16.12.2	Heavy wetting test		
16.12.3	Hydraulic test		
16.12.4	Cantilever load test		
16.13	Tests for SF6 switchgear cable terminations		
16.13.1	General		
16.13.2	Cantilever load test		
16.13.3	Mechanical endurance test		